

# CONTROL CHECK AI

ERD & Database Specification v0.2

PostgreSQL canonical data model | MVP

## 1. Architecture Principle

- Organization-first multi-tenancy; project-scoped analytics.
- Keep raw source rows so every finding can be traced back to the uploaded file and row.
- Normalize imported Excel data into canonical project-control facts.
- Rules and KPI calculations are deterministic backend logic; LLM is reasoning/explanation only.
- Every AI answer should be able to reference finding/evidence records.

## 2. ERD Overview

organizations

■■■ organization\_members ■■ users

■■■ projects

■■■ project\_members ■■ users

■■■ source\_files ■■ datasets ■■ import\_batches ■■ raw\_rows

■■■ wbs\_nodes

■■■ budget\_records

■■■ cost\_records

■■■ schedule\_activities

■■■ progress\_records

■■■ control\_rule\_runs ■■ control\_rules

■■■ findings ■■ finding\_evidence ■■ raw\_rows / canonical facts

■■■ actions ■■ users

■■■ health\_snapshots

■■■ ai\_conversations ■■ ai\_messages

■■■ reports

■■■ audit\_logs

### 3. Core Tables

| organizations          | Tenant/company           |
|------------------------|--------------------------|
| users                  | Identity                 |
| organization_members   | Tenant role              |
| projects               | Project master           |
| project_members        | Project access           |
| source_files           | Uploaded file metadata   |
| datasets               | Logical dataset          |
| import_batches         | Import execution         |
| import_column_mappings | Source→canonical mapping |
| raw_rows               | Immutable source row     |
| wbs_nodes              | WBS hierarchy            |
| budget_records         | Budget facts             |
| cost_records           | Actual/commitment facts  |
| schedule_activities    | Activity facts           |
| progress_records       | Period progress facts    |
| control_rules          | Rule definitions         |
| control_rule_runs      | Rule execution           |
| findings               | Detected issue           |
| finding_evidence       | Evidence links           |
| actions                | Follow-up                |
| health_snapshots       | Health score history     |
| ai_conversations       | AI chat session          |
| ai_messages            | AI/tool interaction      |
| reports                | Generated reports        |
| audit_logs             | Audit trail              |

### 4. Canonical Fact Fields

| Fact type | Canonical fact fields                                                                                                                                                            |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Budget    | project_id, wbs_code, cost_code, period, budget_amount, approved_budget_amount, currency                                                                                         |
| Cost      | transaction_date, wbs_code, cost_code, vendor_code, vendor_name, po_number, document_number, amount, cost_type, currency                                                         |
| Schedule  | activity_code, wbs_code, activity_name, baseline_start, baseline_finish, actual_start, actual_finish, planned_progress, actual_progress, total_float_days, critical_flag, status |
| Progress  | period, wbs_code, planned_progress, actual_progress, progress_method                                                                                                             |

## 5. Detailed Table Specification

### organizations

| column     | type         | constraint / meaning |
|------------|--------------|----------------------|
| id         | uuid         | PK                   |
| name       | varchar(200) | required             |
| slug       | varchar(100) | unique               |
| status     | varchar(20)  | active suspended     |
| created_at | timestampz   |                      |
| updated_at | timestampz   |                      |

### users

| column        | type         | constraint / meaning    |
|---------------|--------------|-------------------------|
| id            | uuid         | PK                      |
| email         | varchar(320) | unique                  |
| full_name     | varchar(200) | required                |
| password_hash | text         | nullable                |
| status        | varchar(20)  | active invited disabled |
| created_at    | timestampz   |                         |
| updated_at    | timestampz   |                         |

### organization\_members

| column          | type        | constraint / meaning               |
|-----------------|-------------|------------------------------------|
| organization_id | uuid        | PK/FK                              |
| user_id         | uuid        | PK/FK                              |
| role            | varchar(30) | owner admin manager analyst viewer |
| joined_at       | timestampz  |                                    |
| is_active       | boolean     | default true                       |

### projects

| column          | type         | constraint / meaning                     |
|-----------------|--------------|------------------------------------------|
| id              | uuid         | PK                                       |
| organization_id | uuid         | FK                                       |
| code            | varchar(80)  | unique per org                           |
| name            | varchar(250) | required                                 |
| client_name     | varchar(250) | nullable                                 |
| project_type    | varchar(80)  | EPC construction oil_gas mining other    |
| currency        | char(3)      | ISO-4217                                 |
| planned_start   | date         | nullable                                 |
| planned_finish  | date         | nullable                                 |
| status          | varchar(30)  | planning active on_hold completed closed |
| created_at      | timestampz   |                                          |
| updated_at      | timestampz   |                                          |

## project\_members

| Column     | Type        | Constraint / Meaning   |
|------------|-------------|------------------------|
| project_id | uuid        | PK/FK                  |
| user_id    | uuid        | PK/FK                  |
| role       | varchar(30) | manager analyst viewer |
| is_active  | boolean     | default true           |

## source\_files

| Column          | Type         | Constraint / Meaning |
|-----------------|--------------|----------------------|
| id              | uuid         | PK                   |
| organization_id | uuid         | FK                   |
| project_id      | uuid         | FK                   |
| file_name       | varchar(255) | required             |
| storage_key     | text         | required             |
| mime_type       | varchar(100) |                      |
| file_size_bytes | bigint       |                      |
| sha256          | char(64)     |                      |
| uploaded_by     | uuid         | FK users             |
| uploaded_at     | timestampz   |                      |

## datasets

| Column         | Type         | Constraint / Meaning                                  |
|----------------|--------------|-------------------------------------------------------|
| id             | uuid         | PK                                                    |
| project_id     | uuid         | FK                                                    |
| source_file_id | uuid         | FK nullable                                           |
| dataset_type   | varchar(30)  | budget actual_cost schedule progress commitment other |
| name           | varchar(150) |                                                       |
| schema_version | varchar(20)  |                                                       |
| status         | varchar(20)  | draft mapped validated imported failed                |
| record_count   | integer      |                                                       |
| created_at     | timestampz   |                                                       |

## import\_batches

| Column        | Type        | Constraint / Meaning           |
|---------------|-------------|--------------------------------|
| id            | uuid        | PK                             |
| dataset_id    | uuid        | FK                             |
| started_at    | timestampz  |                                |
| completed_at  | timestampz  | nullable                       |
| status        | varchar(20) | running success partial failed |
| rows_read     | integer     |                                |
| rows_valid    | integer     |                                |
| rows_rejected | integer     |                                |
| error_summary | jsonb       |                                |

## import\_column\_mappings

| Column          | Type         | Constraint / Meaning                |
|-----------------|--------------|-------------------------------------|
| id              | uuid         | PK                                  |
| dataset_id      | uuid         | FK                                  |
| source_column   | varchar(200) |                                     |
| canonical_field | varchar(100) |                                     |
| data_type       | varchar(30)  | string date decimal integer boolean |
| mapping_method  | varchar(20)  | manual rule ai                      |
| confidence      | numeric(5,4) | 0..1                                |
| is_confirmed    | boolean      |                                     |

## raw\_rows

| Column            | Type        | Constraint / Meaning   |
|-------------------|-------------|------------------------|
| id                | bigserial   | PK                     |
| import_batch_id   | uuid        | FK                     |
| source_row_number | integer     |                        |
| row_hash          | char(64)    |                        |
| raw_data          | jsonb       |                        |
| validation_status | varchar(20) | valid warning rejected |
| validation_errors | jsonb       | nullable               |

## wbs\_nodes

| Column     | Type         | Constraint / Meaning |
|------------|--------------|----------------------|
| id         | uuid         | PK                   |
| project_id | uuid         | FK                   |
| parent_id  | uuid         | self FK nullable     |
| wbs_code   | varchar(100) | unique per project   |
| wbs_name   | varchar(250) | nullable             |
| level_no   | integer      | >=1                  |
| path       | ltree        | optional             |
| is_active  | boolean      |                      |

## budget\_records

| Column                 | Type          | Constraint / Meaning |
|------------------------|---------------|----------------------|
| id                     | uuid          | PK                   |
| project_id             | uuid          | FK                   |
| wbs_node_id            | uuid          | FK nullable          |
| dataset_id             | uuid          | FK                   |
| raw_row_id             | bigint        | FK nullable          |
| cost_code              | varchar(100)  | nullable             |
| period                 | date          | nullable             |
| budget_amount          | numeric(20,2) | >=0                  |
| approved_budget_amount | numeric(20,2) | nullable             |
| currency               | char(3)       |                      |
| created_at             | timestampz    |                      |

## cost\_records

| Column           | Type          | Constraint / Meaning                    |
|------------------|---------------|-----------------------------------------|
| id               | uuid          | PK                                      |
| project_id       | uuid          | FK                                      |
| wbs_node_id      | uuid          | FK nullable                             |
| dataset_id       | uuid          | FK                                      |
| raw_row_id       | bigint        | FK nullable                             |
| transaction_date | date          |                                         |
| cost_code        | varchar(100)  | nullable                                |
| vendor_name      | varchar(250)  | nullable                                |
| vendor_code      | varchar(100)  | nullable                                |
| po_number        | varchar(100)  | nullable                                |
| document_number  | varchar(100)  | nullable                                |
| description      | text          | nullable                                |
| amount           | numeric(20,2) | >=0                                     |
| cost_type        | varchar(30)   | actual commitment accrual invoice other |
| currency         | char(3)       |                                         |

## schedule\_activities

| column           | type         | constraint / meaning                                |
|------------------|--------------|-----------------------------------------------------|
| id               | uuid         | PK                                                  |
| project_id       | uuid         | FK                                                  |
| wbs_node_id      | uuid         | FK nullable                                         |
| dataset_id       | uuid         | FK                                                  |
| raw_row_id       | bigint       | FK nullable                                         |
| activity_code    | varchar(100) |                                                     |
| activity_name    | varchar(500) |                                                     |
| baseline_start   | date         | nullable                                            |
| baseline_finish  | date         | nullable                                            |
| actual_start     | date         | nullable                                            |
| actual_finish    | date         | nullable                                            |
| planned_progress | numeric(7,4) | 0..1                                                |
| actual_progress  | numeric(7,4) | 0..1                                                |
| total_float_days | integer      | nullable                                            |
| critical_flag    | boolean      | nullable                                            |
| status           | varchar(30)  | not_started in_progress completed delayed cancelled |

## progress\_records

| column           | type         | constraint / meaning |
|------------------|--------------|----------------------|
| id               | uuid         | PK                   |
| project_id       | uuid         | FK                   |
| wbs_node_id      | uuid         | FK nullable          |
| dataset_id       | uuid         | FK                   |
| raw_row_id       | bigint       | FK nullable          |
| period           | date         |                      |
| planned_progress | numeric(7,4) | 0..1                 |
| actual_progress  | numeric(7,4) | 0..1                 |
| progress_method  | varchar(50)  |                      |

## control\_rules

| Column               | Type         | Constraint / Meaning                                 |
|----------------------|--------------|------------------------------------------------------|
| id                   | uuid         | PK                                                   |
| code                 | varchar(80)  | unique                                               |
| name                 | varchar(200) |                                                      |
| category             | varchar(30)  | data_quality cost schedule progress cross_d<br>omain |
| severity_default     | varchar(20)  | critical warning observation                         |
| version              | integer      |                                                      |
| expression           | jsonb        | declarative                                          |
| explanation_template | text         | nullable                                             |
| is_active            | boolean      |                                                      |
| created_at           | timestampz   |                                                      |

## control\_rule\_runs

| Column           | Type        | Constraint / Meaning   |
|------------------|-------------|------------------------|
| id               | uuid        | PK                     |
| project_id       | uuid        | FK                     |
| rule_id          | uuid        | FK                     |
| started_at       | timestampz  |                        |
| completed_at     | timestampz  | nullable               |
| status           | varchar(20) | running success failed |
| records_scanned  | integer     |                        |
| findings_created | integer     |                        |
| execution_ms     | integer     | nullable               |



## findings

| column          | type          | constraint / meaning                             |
|-----------------|---------------|--------------------------------------------------|
| id              | uuid          | PK                                               |
| project_id      | uuid          | FK                                               |
| rule_run_id     | uuid          | FK nullable                                      |
| rule_id         | uuid          | FK nullable                                      |
| category        | varchar(30)   |                                                  |
| severity        | varchar(20)   | critical warning observation                     |
| status          | varchar(20)   | open acknowledged in_progress resolved dismissed |
| title           | varchar(300)  |                                                  |
| description     | text          |                                                  |
| metric_json     | jsonb         |                                                  |
| impact_amount   | numeric(20,2) | nullable                                         |
| confidence      | numeric(5,4)  | 0..1                                             |
| wbs_node_id     | uuid          | FK nullable                                      |
| detected_at     | timestampz    |                                                  |
| resolved_at     | timestampz    | nullable                                         |
| resolution_note | text          | nullable                                         |

## finding\_evidence

| column        | type        | constraint / meaning              |
|---------------|-------------|-----------------------------------|
| id            | uuid        | PK                                |
| finding_id    | uuid        | FK                                |
| raw_row_id    | bigint      | FK nullable                       |
| entity_type   | varchar(50) | budget cost schedule progress wbs |
| entity_id     | uuid        | nullable                          |
| evidence_role | varchar(30) | primary supporting context        |
| excerpt       | jsonb       | nullable                          |

## actions

| column        | type         | constraint / meaning            |
|---------------|--------------|---------------------------------|
| id            | uuid         | PK                              |
| project_id    | uuid         | FK                              |
| finding_id    | uuid         | FK nullable                     |
| title         | varchar(300) |                                 |
| description   | text         | nullable                        |
| priority      | varchar(20)  | high medium low                 |
| owner_user_id | uuid         | FK users nullable               |
| due_date      | date         | nullable                        |
| status        | varchar(20)  | open in_progress done cancelled |
| created_at    | timestampz   |                                 |
| completed_at  | timestampz   | nullable                        |

## health\_snapshots

| column             | type         | constraint / meaning |
|--------------------|--------------|----------------------|
| id                 | uuid         | PK                   |
| project_id         | uuid         | FK                   |
| snapshot_date      | date         |                      |
| overall_score      | numeric(6,2) | 0..100               |
| cost_score         | numeric(6,2) | 0..100               |
| schedule_score     | numeric(6,2) | 0..100               |
| progress_score     | numeric(6,2) | 0..100               |
| data_quality_score | numeric(6,2) | 0..100               |
| score_version      | varchar(20)  |                      |
| details            | jsonb        | nullable             |

## ai\_conversations

| column          | type         | constraint / meaning |
|-----------------|--------------|----------------------|
| id              | uuid         | PK                   |
| organization_id | uuid         | FK                   |
| project_id      | uuid         | FK nullable          |
| user_id         | uuid         | FK                   |
| title           | varchar(250) | nullable             |
| created_at      | timestampz   |                      |
| updated_at      | timestampz   |                      |

## ai\_messages

| column          | type         | constraint / meaning       |
|-----------------|--------------|----------------------------|
| id              | uuid         | PK                         |
| conversation_id | uuid         | FK                         |
| role            | varchar(20)  | user assistant system tool |
| content         | text         | nullable                   |
| tool_name       | varchar(100) | nullable                   |
| tool_payload    | jsonb        | nullable                   |
| evidence_refs   | jsonb        | nullable                   |
| model           | varchar(100) | nullable                   |
| input_tokens    | integer      | nullable                   |
| output_tokens   | integer      | nullable                   |
| created_at      | timestampz   |                            |

## reports

| Column       | Type        | Constraint / Meaning                           |
|--------------|-------------|------------------------------------------------|
| id           | uuid        | PK                                             |
| project_id   | uuid        | FK                                             |
| requested_by | uuid        | FK users                                       |
| report_type  | varchar(30) | monthly executive findings data_quality custom |
| period_start | date        | nullable                                       |
| period_end   | date        | nullable                                       |
| status       | varchar(20) | queued generating ready failed                 |
| storage_key  | text        | nullable                                       |
| created_at   | timestampz  |                                                |
| completed_at | timestampz  | nullable                                       |

## audit\_logs

| Column          | Type         | Constraint / Meaning |
|-----------------|--------------|----------------------|
| id              | bigserial    | PK                   |
| organization_id | uuid         | FK                   |
| user_id         | uuid         | FK nullable          |
| project_id      | uuid         | FK nullable          |
| event_type      | varchar(100) |                      |
| entity_type     | varchar(80)  | nullable             |
| entity_id       | varchar(100) | nullable             |
| metadata        | jsonb        | nullable             |
| ip_address      | inet         | nullable             |
| created_at      | timestampz   |                      |

## 6. Key Constraints

- Project code unique within organization; WBS code unique within project.
- Progress values constrained to 0..1.
- Every normalized fact carries dataset\_id and, where available, raw\_row\_id.
- Finding evidence is created with findings in one transaction.
- Cross-tenant access is rejected at the service layer and later reinforced with PostgreSQL RLS.
- Use UUID keys for business entities; BIGSERIAL for high-volume raw/audit rows.

## 7. Index Strategy

| Index    | Fields                                                                                               |
|----------|------------------------------------------------------------------------------------------------------|
| Projects | organization_id, status                                                                              |
| Costs    | project_id+transaction_date; project_id+wbs_node_id;<br>project_id+po_number; project_id+vendor_code |
| Schedule | project_id+baseline_finish; project_id+critical_flag; project_id+status                              |
| Findings | project_id+status+severity; project_id+category;<br>project_id+detected_at                           |
| Evidence | finding_id; raw_row_id                                                                               |
| Audit    | organization_id+created_at; project_id+created_at                                                    |

## 8. MVP Seed Rules

| Rule ID | Rule Description                             | Severity |
|---------|----------------------------------------------|----------|
| DQ-001  | Missing WBS in cost data                     | Warning  |
| DQ-002  | Duplicate transaction/document               | Warning  |
| DQ-003  | Invalid date / finish before start           | Critical |
| DQ-004  | Progress outside 0..1                        | Critical |
| CST-001 | Actual cumulative cost > approved budget     | Critical |
| CST-002 | Actual cost > WBS budget                     | Warning  |
| CST-003 | Cost acceleration spike                      | Warning  |
| CST-004 | Actual + commitment > budget                 | Critical |
| SCH-001 | Activity past baseline finish and incomplete | Warning  |
| SCH-002 | Critical activity delayed                    | Critical |
| SCH-003 | Actual progress behind planned               | Warning  |
| SCH-004 | Negative float below threshold               | Critical |
| PRG-001 | Progress behind plan                         | Warning  |
| XDM-001 | Cost rising while progress remains flat      | Critical |

## 9. Health Score v0.1

Overall = 0.30 Cost + 0.30 Schedule + 0.25 Progress + 0.15 Data Quality. Category penalties are driven by validated finding severity and recurrence. The scoring curve is configuration-based and versioned.

| Score Range | Risk Category |
|-------------|---------------|
| 90–100      | Healthy       |
| 75–89.99    | Watch         |
| 60–74.99    | Moderate Risk |
| 40–59.99    | High Risk     |
| 0–39.99     | Critical Risk |

## 10. API Boundary

| Resource | API Endpoints                                                                          |
|----------|----------------------------------------------------------------------------------------|
| Projects | GET/POST /projects; GET/PATCH /projects/{id}                                           |
| Datasets | POST /projects/{id}/datasets; POST /datasets/{id}/validate; POST /datasets/{id}/import |
| Findings | GET /projects/{id}/findings; GET/PATCH /findings/{id}; GET /findings/{id}/evidence     |
| Rules    | GET /control-rules; POST /projects/{id}/control-runs                                   |
| Health   | GET /projects/{id}/health/current; GET /projects/{id}/health/history                   |
| Actions  | POST /findings/{id}/actions; PATCH /actions/{id}                                       |
| AI       | POST /projects/{id}/ai/conversations; POST /conversations/{id}/messages                |
| Reports  | POST /projects/{id}/reports; GET /reports/{id}                                         |

## 11. Build Order

| Order | Task                                  |
|-------|---------------------------------------|
| 1     | PostgreSQL migrations + seed rules    |
| 2     | Auth / organization / project tenancy |
| 3     | File upload + source file metadata    |
| 4     | Dataset mapping + raw rows            |
| 5     | Canonical importers                   |
| 6     | WBS normalization                     |
| 7     | Control engine                        |
| 8     | Finding + evidence engine             |
| 9     | Health scoring                        |
| 10    | FastAPI                               |
| 11    | React screens                         |
| 12    | AI reasoning + assistant              |

## 12. Definition of Done

- Migrations run cleanly on empty PostgreSQL.
- Sample project can ingest four datasets.
- At least 10 seed rules execute and create findings.
- Every finding has traceable evidence.
- Cross-tenant access tests fail closed.
- Health snapshot is reproducible from the same input data and score version.
- API contract is documented and versioned.

## Phase 4A Persistence Alignment

Version 0.2 aligns the conceptual ERD with the executable PostgreSQL persistence layer delivered in Phase 4A. Alembic migration 20260817\_0001 is the executable schema authority; the SQL reference and this document are readable specifications.

## Implemented Phase 4A tables

| organizations           | Tenant boundary and lifecycle status      | Parent of all tenant-scoped records                  |
|-------------------------|-------------------------------------------|------------------------------------------------------|
| projects                | Project identity and planning envelope    | Unique organization + project code                   |
| source_files            | Immutable workbook metadata and SHA-256   | Project-scoped storage reference                     |
| dataset_snapshots       | Validated analysis input at a data date   | Connects project, source file, and source project ID |
| rule_catalogue_versions | Versioned JSONB rule definitions          | Version + SHA-256 unique identity                    |
| analysis_runs           | Durable engine execution lifecycle        | queued/running/succeeded/failed                      |
| findings                | Materialized deterministic findings       | Unique engine finding per analysis run               |
| finding_evidence        | Ordered, source-traceable evidence        | Rows, record IDs, fields, aggregation in JSONB       |
| approved_exceptions     | Governed suppressions and validity window | Execution support is scheduled after Phase 4A        |
| audit_logs              | Append-oriented lifecycle event history   | Records status changes and tenant context            |

## Contract decisions

- All tenant-sensitive queries include organization\_id in the database predicate.
- Project workbook identity must match the registered project code before a durable run is created.
- findings.entity\_id is text (up to 300 characters), because deterministic engine entities include composite identifiers—not only UUIDs.
- A failed engine execution retains an analysis\_runs record with a safe error code and message, but creates no findings.
- finding\_evidence preserves ordered JSONB source references for auditability while raw and canonical fact tables remain Phase 4B scope.
- Authentication and complete RBAC are deferred; Phase 4A uses X-Organization-ID as a temporary tenant-context contract.

## Phase boundary

Phase 4A persists project metadata, uploaded file metadata, dataset snapshots, catalogue versions, analysis runs, findings, evidence, exception definitions, and audit events. Phase 4B adds raw row lineage and normalized WBS, budget, cost, commitment, schedule, and progress fact tables.